

B - Sensor Data Logging

分值: 200 分

Problem Statement

In a certain measurement, the sensor readings at times $0, 1, \dots, T$ are recorded according to the following rules.

在某次测量中, 时刻 $0, 1, \dots, T$ 的传感器读数按照以下规则记录。

- At time 0, the reading is saved.
 - At times $1, 2, \dots, T$, the reading is saved if and only if the absolute difference between the current reading and the most recently saved reading is at least X .
 - 在时刻 0, 读数一定会被保存。
 - 对于时刻 $1, 2, \dots, T$, 当且仅当当前读数与最近一次保存的读数的绝对差至少为 X 时, 该读数才会被保存。
-

The sensor reading at time $i = 0, 1, \dots, T$ was A_i .

时刻 $i = 0, 1, \dots, T$ 的传感器读数为 A_i 。

Output the times at which readings were saved and the saved values, in ascending order of time.

按时间升序输出所有保存了读数的时刻, 以及对应的保存值。

Constraints

- $1 \leq T \leq 100$
- $1 \leq X \leq 100$
- $0 \leq A_i \leq 100$
- All input values are integers.
- $1 \leq T \leq 100$
- $1 \leq X \leq 100$

- $0 \leq A_i \leq 100$
- 所有输入值均为整数。

Input

The input is given from Standard Input in the following format:

输入从标准输入按照以下格式给出：

T X
 A_0 A_1 $\dots$ A_T

Output

If k readings were saved, and the i -th saved reading in ascending order of time was at time t_i with value a_i , output in the following format:

假设共保存了 k 个读数，其中按时间升序的第 i 个读数的保存时刻为 t_i 、数值为 a_i ，请按照以下格式输出：

```
$t_1$ $a_1$  
$t_2$ $a_2$  
$\vdots$  
$t_k$ $a_k$
```

```
$t_1$ $a_1$  
$t_2$ $a_2$  
$\vdots$  
$t_k$ $a_k$
```

Sample Input 1

```
6 10  
30 35 40 21 30 12 31
```

Sample Output 1

```
0 30
2 40
3 21
6 31
```

The measurement proceeds as follows.

测量过程如下。

- The reading at time 0 is 30. Save it.
- The reading at time 1 is 35. The most recently saved reading is 30, and the absolute difference is less than 10, so it is not saved.
- The reading at time 2 is 40. The most recently saved reading is 30, and the absolute difference is at least 10, so it is saved.
- The reading at time 3 is 21. The most recently saved reading is 40, and the absolute difference is at least 10, so it is saved.
- The reading at time 4 is 30. The most recently saved reading is 21, and the absolute difference is less than 10, so it is not saved.
- The reading at time 5 is 12. The most recently saved reading is 21, and the absolute difference is less than 10, so it is not saved.
- The reading at time 6 is 31. The most recently saved reading is 21, and the absolute difference is at least 10, so it is saved.
- 时刻 0 的读数为 30，保存该读数。
- 时刻 1 的读数为 35，最近一次保存的读数为 30，两者绝对差小于 10，因此不保存。
- 时刻 2 的读数为 40，最近一次保存的读数为 30，两者绝对差至少为 10，因此保存该读数。
- 时刻 3 的读数为 21，最近一次保存的读数为 40，两者绝对差至少为 10，因此保存该读数。
- 时刻 4 的读数为 30，最近一次保存的读数为 21，两者绝对差小于 10，因此不保存。
- 时刻 5 的读数为 12，最近一次保存的读数为 21，两者绝对差小于 10，因此不保存。
- 时刻 6 的读数为 31，最近一次保存的读数为 21，两者绝对差至少为 10，因此保存该读数。